Social Media Engagement Analysis

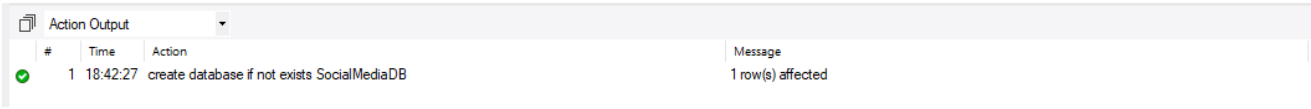
Tasks Performed:

Tasks 1:

DDL (CREATE, ALTER, DROP, RENAME, TRUNCATE)

Challenge 1 — Create Database

Create a database named SocialMediaDB



Challenge 2 — Create Table

Create a appusers table:

Column Type

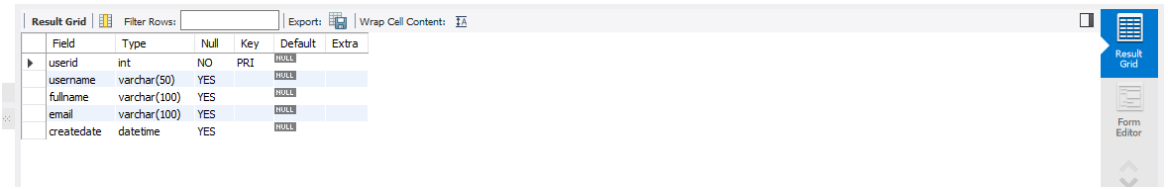
userid INT

username VARCHAR(50)

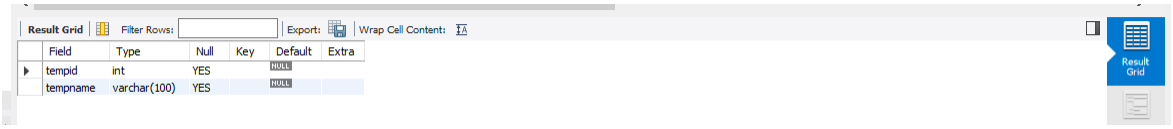
fullname VARCHAR(100)

email VARCHAR(100)

createdat DATETIME

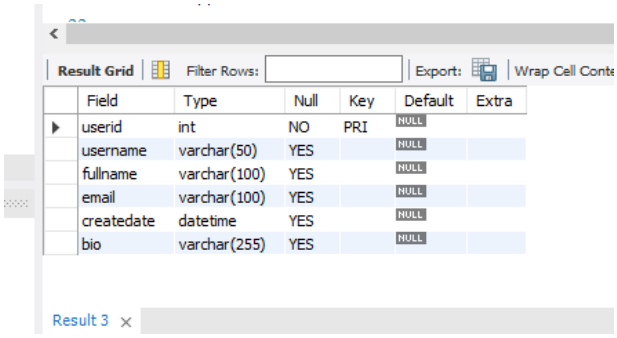


• create a temptable --tempid int datatype,tempname varchar(100)



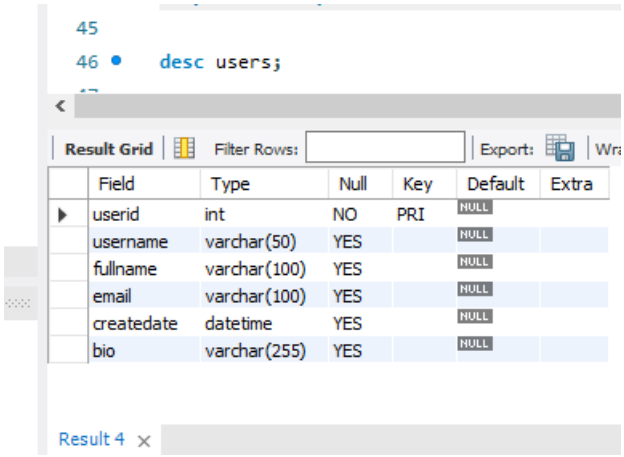
Challenge 3 — Alter Table

Add column bio (VARCHAR 255) to users.



Challenge 4 — Rename & Drop

Rename table appusers → users.



Drop a table named temp_table if it exists.

```
43
44 •  drop table temptable;
45 •  desc temptable;
46
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:52:14	create database if not exists SocialMediaDB	1 row(s) affected	0.016 sec
2	18:52:16	use SocialMediaDB	0 row(s) affected	0.000 sec
3	18:52:20	create table if not exists appusers(userid int primary key, username varchar(50), fullname varch...	0 row(s) affected	0.047 sec
4	18:52:28	desc appusers	5 row(s) returned	0.000 sec / 0.000 sec
5	18:52:35	create table if not exists temptable(tempid int, tempname varchar(100))	0 row(s) affected	0.047 sec
6	18:52:41	desc temptable	2 row(s) returned	0.000 sec / 0.000 sec
7	18:53:01	alter table appusers add column bio varchar(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
8	18:53:07	desc appusers	6 row(s) returned	0.000 sec / 0.000 sec
9	18:54:03	rename table appusers to users	0 row(s) affected	0.032 sec
10	18:54:22	desc users	6 row(s) returned	0.000 sec / 0.000 sec
11	18:55:10	drop table temptable	0 row(s) affected	0.015 sec
12	18:55:36	desc temptable	Error Code: 1146. Table 'socialmediadb.temptable' doesn't exist	0.000 sec

Tasks 2:

KEYS, CONSTRAINTS, INSERT/UPDATE/DELETE

Challenge 1 — Posts Table with Constraints

Create a posts table:

Column -Type -Constraint

post_id-INT- PK, autoincrement

**userid*- INT -FK → users.user_id

caption-TEXT

postedat-DATETIME ->DEFAULT CURRENT-TIMESTAMP

```
68 •  desc Posts;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
postid	int	NO	PRI	NULL	auto_increment
userid	int	YES	MUL	NULL	
caption	text	YES		NULL	
posted_at	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

Challenge 2 – Followers Table

Create a followers table:

Column-Type-->Constraint

*followerid-*INT-->FK → users.user_id

*followingid**-INT-->FK → users.userid

follow_date-DATETIME-->DEFAULT CURRENT_TIMESTAMP

```
84
85 •        desc Followers;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
followerid	int	YES	MUL	NULL	
followingid	int	YES	MUL	NULL	
follow_date	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

Result 6 ×

Output

Challenge 3 – Insert Sample Data

Insert 5 users and 5 posts.

```
105
106 •   Select * from users;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cor

userid	username	fullname	email	created_at	bio
101	Kala	Kala Ram	kala@gmail.com	2023-01-23 10:30:00	Cooking Blogger
102	Mala	Mala Vel	mala@yahoo.com	2024-02-05 10:30:00	Yoga Trainer
103	Varun	Varun Raj	varun@hotmail.com	2025-03-15 10:30:00	Singer
104	Sri	Sri Ram	sri@gmail.com	2026-04-26 10:30:00	Health Coach
105	Mathu	Mathu Sri	Mathu@hotmail.com	2026-05-25 10:30:00	Montessori Teacher
NULL	NULL	NULL	NULL	NULL	NULL

users 7 ×

Output

```
107 •   Select * from posts;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

postid	userid	caption	posted_at
1	101	Creating memorable meals	2023-01-24 11:00:00
2	102	NULL	2024-02-06 11:00:00
3	103	Every song is a piece of my heart	2025-03-16 11:00:00
4	104	Healthy habits, happy life	2026-04-27 11:00:00
5	105	NULL	2026-05-26 11:00:00
NULL	NULL	NULL	NULL

posts 8 ×

Output

Task 3:

DISTINCT, WHERE, IN, BETWEEN, NULL checks

Challenge 1 — Filter Users

Retrieve all users whose username starts with “m”

```
114
115 •  Select * from users where username like ('m%');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

userid	username	fullname	email	created_at	bio
102	Mala	Mala Vel	mala@yahoo.com	2024-02-05 10:30:00	Yoga Trainer
105	Mathu	Mathu Sri	Mathu@hotmail.com	2026-05-25 10:30:00	Montessori Teacher
NULL	NULL	NULL	NULL	NULL	NULL

users 9 ×

Output

Challenge 2 — Posts Filter

Get posts posted BETWEEN '2023-01-01' AND '2025-12-31'

```
119
120 •  Select * from posts where posted_at between '2023-01-01'
```

Result Grid | Filter Rows: | Edit: | Export/Import

postid	userid	caption	posted_at
1	101	Creating memorable meals	2023-01-24 11:00:00
2	102	NULL	2024-02-06 11:00:00
3	103	Every song is a piece of my heart	2025-03-16 11:00:00
NULL	NULL	NULL	NULL

posts 10 ×

Challenge 3 — NULL checks

Find posts that have NULL captions.

```
124
125 •  Select * from posts where caption IS NULL;
```

Result Grid | Filter Rows: | Edit: | Expo

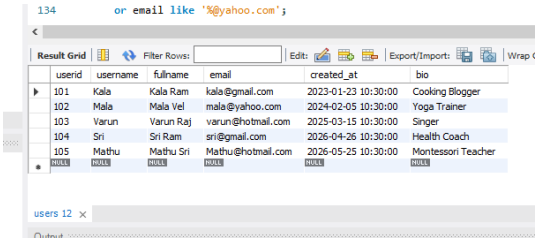
postid	userid	caption	posted_at
2	102	NULL	2024-02-06 11:00:00
5	105	NULL	2026-05-26 11:00:00
NULL	NULL	NULL	NULL

posts 11 ×

Challenge 4 — IN Clause

Find users whose email domain is in:

(@gmail.com, @hotmail.com, @yahoo.com)



```
134        or email like '%@yahoo.com';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

userid	username	fullname	email	created_at	bio
101	Kala	Kala Ram	kala@gmail.com	2023-01-23 10:30:00	Cooking Blogger
102	Mala	Mala Vel	mala@yahoo.com	2024-02-05 10:30:00	Yoga Trainer
103	Varun	Varun Raj	varun@hotmail.com	2025-03-15 10:30:00	Singer
104	Sri	Sri Ram	sri@gmail.com	2026-04-26 10:30:00	Health Coach
105	Mathu	Mathu Sri	Mathu@hotmail.com	2026-05-25 10:30:00	Montessori Teacher
NULL	NULL	NULL	NULL	NULL	NULL

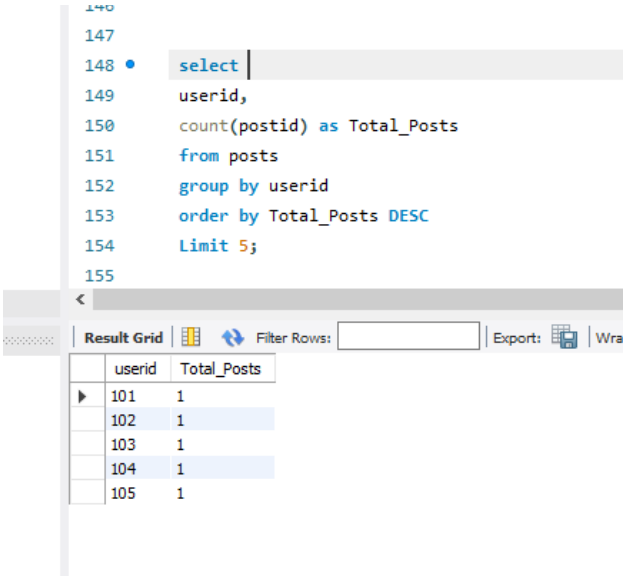
users 12

Task 4:

ORDER BY, GROUP BY, LIMIT, HAVING

Challenge 1 — Top Active Users

• Sort users by number of posts (DESC) and LIMIT 5.



```
147
148 •  select
149    userid,
150    count(postid) as Total_Posts
151    from posts
152    group by userid
153    order by Total_Posts DESC
154    Limit 5;
155
```

Result Grid | Filter Rows: | Export: | Wra

userid	Total_Posts
101	1
102	1
103	1
104	1
105	1

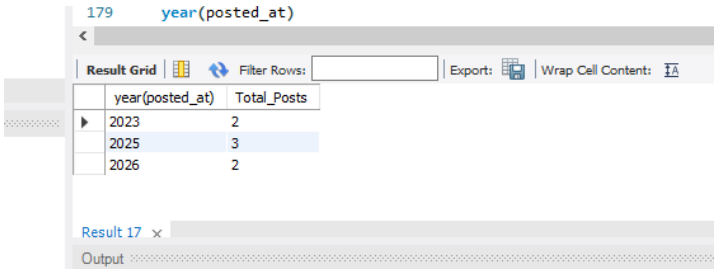
• Challenge 2 — Group by Posts Per Day

Group by Posts Per YEAR and return:

· date

· total posts

Only show days with more than 10 posts (HAVING).



Task 5:

JOINS, UNION, BUxILT-IN FUNCTIONS

Challenge 1 — Comments Table

Create:

comments

· commentid (PK)

· postid (FK)

· userid (FK)

· commenttext

· commentedat

likes

· likeid (PK)

· postid (FK)

· userid (FK)

· likedat

```
246
247 •  select * from comments;
248
249 •  INSERT INTO likes (postid, userid, liked_at)
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Ce

commentid	postid	userid	commenttext	commented_at
1	1	102	Looks delicious!	2023-01-24 14:00:00
2	1	103	Amazing cooking!	2023-01-24 14:30:00
3	2	101	Nice post!	2024-02-06 14:00:00
4	3	102	Beautiful song!	2025-03-16 14:00:00
5	3	104	Your voice is amazing!	2025-03-16 14:30:00
6	3	105	Loved this song!	2025-03-16 15:00:00
7	4	101	Great health tips!	2026-04-27 14:00:00
8	4	103	Very useful information!	2026-04-27 14:30:00
9	5	101	Interesting post!	2026-05-26 14:00:00
10	6	102	The food looks tasty!	2023-01-25 14:00:00
11	6	105	Great recipe!	2023-01-25 14:30:00
12	7	104	Beautiful melody!	2025-03-17 14:00:00
13	7	105	I love this song!	2025-03-17 14:30:00
14	8	101	AR Rahman songs are ...	2025-03-18 14:00:00
15	8	102	My favorite music!	2025-03-18 14:30:00
16	8	105	Great collection!	2025-03-18 15:00:00
NULL	NULL	NULL	NULL	NULL

comments 18 ×

```
278
279 •  select * from likes;
```

Result Grid | Filter Rows: | Edit:

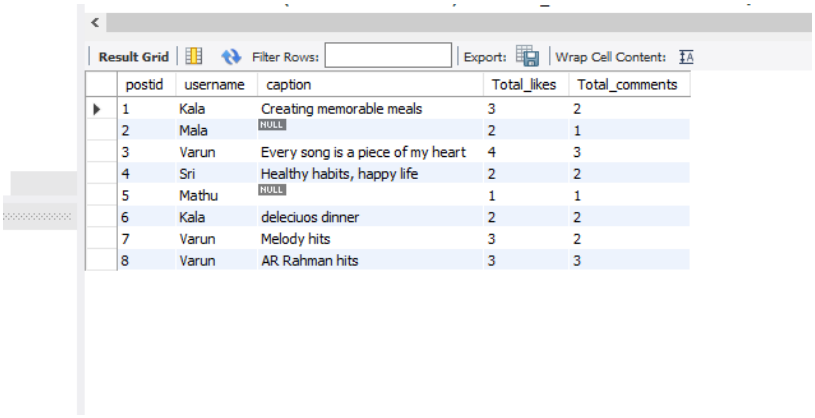
likeid	postid	userid	liked_at
1	1	102	2023-01-24 12:00:00
2	1	103	2023-01-24 12:30:00
3	1	104	2023-01-24 13:00:00
4	2	101	2024-02-06 12:00:00
5	2	103	2024-02-06 12:30:00
6	3	101	2025-03-16 12:00:00
7	3	102	2025-03-16 12:30:00
8	3	104	2025-03-16 13:00:00
9	3	105	2025-03-16 13:30:00
10	4	101	2026-04-27 12:00:00
11	4	103	2026-04-27 12:30:00
12	5	102	2026-05-26 12:00:00
13	6	103	2023-01-25 12:00:00
14	6	104	2023-01-25 12:30:00
15	7	101	2025-03-17 12:00:00
16	7	102	2025-03-17 12:30:00
17	7	105	2025-03-17 13:00:00
18	8	101	2025-03-18 12:00:00
19	8	102	2025-03-18 12:30:00
20	8	104	2025-03-18 13:00:00
NULL	NULL	NULL	NULL

likes 19 ×

Challenge 2 — JOIN Report

Display:

postid, username, caption, total likes, total comments.



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

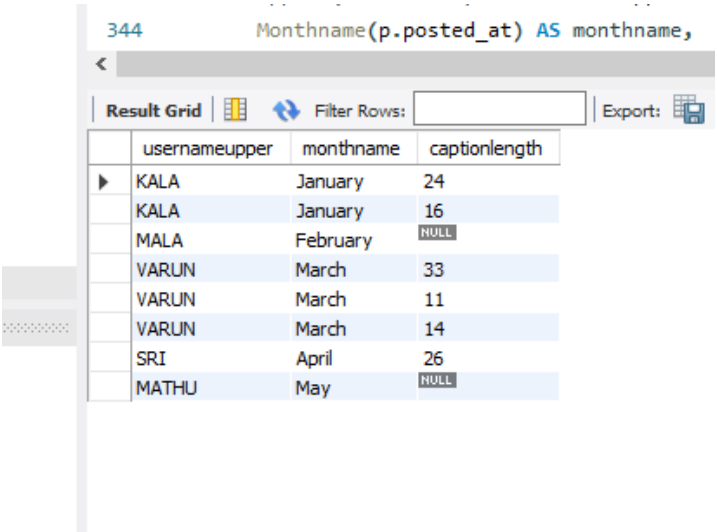
postid	username	caption	Total_likes	Total_comments
1	Kala	Creating memorable meals	3	2
2	Mala	NULL	2	1
3	Varun	Every song is a piece of my heart	4	3
4	Sri	Healthy habits, happy life	2	2
5	Mathu	NULL	1	1
6	Kala	deleciuos dinner	2	2
7	Varun	Melody hits	3	2
8	Varun	AR Rahman hits	3	3

Challenge 3 — Built-in Functions

· Show usernames in UPPER case.

· Extract month name from posted_at.

· Return the length of each caption.



```
344    Monthname(p.posted_at) AS monthname,
```

Result Grid | Filter Rows: | Export:

usernameupper	monthname	captionlength
KALA	January	24
KALA	January	16
MALA	February	NULL
VARUN	March	33
VARUN	March	11
VARUN	March	14
SRI	April	26
MATHU	May	NULL

Challenge 4 — UNION

Combine list of users who commented OR liked.

```
357     JOIN comments c
358         ON u.userid = c.userid
```

Result Grid | Filter Rows: | Export:

userid	username
101	Kala
102	Mala
103	Varun
104	Sri
105	Mathu

Task 6:

FUNCTIONS, SUBQUERIES, STORED PROCEDURES

Challenge 1 — User Defined Function

Create a function:

GetUserEngagement(user_id)

Returns total likes + comments made by the user.

```
399
400 •   -- To call
401
402     select userid, username, GetUserEngagement (userid) as total_likedandcommented_users  from users;
```

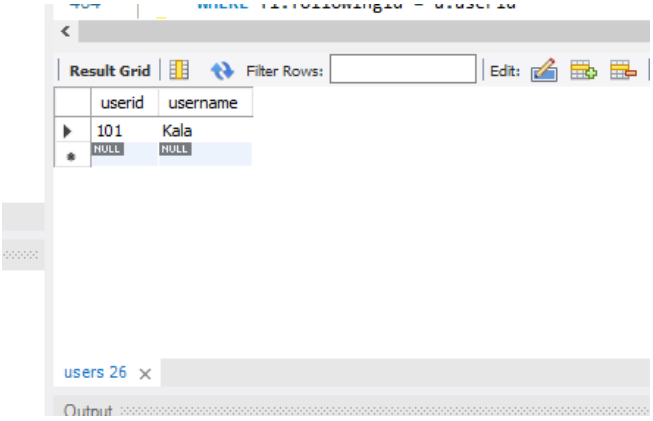
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

userid	username	total_likedandcommented_users
101	Kala	9
102	Mala	9
103	Varun	6
104	Sri	6
105	Mathu	6

Result 25 ×

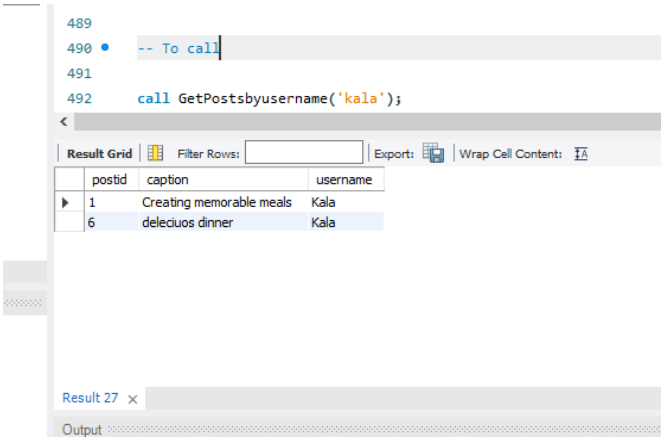
Challenge 2 — Subquery

Find users who have more followers than the average follower count.



Challenge 3 — Stored Procedure

Stored procedure to retrieve all posts for a given username.



Task 7:

VIEWS & TRIGGERS

Challenge 1 — View

Create a view postsummary showing:

· postid

· username

· caption

· total likes

· total comments

```
551
552 •  select * from postsummary2;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PostID	UserName	Caption	Total_Likes	Total_Comments
1	Kala	Creating memorable meals	3	2
2	Mala	NULL	2	1
3	Varun	Every song is a piece of my heart	4	3
4	Sri	Healthy habits, happy life	2	2
5	Mathu	NULL	1	1
6	Kala	deleciuos dinner	2	2
7	Varun	Melody hits	3	2
8	Varun	AR Rahman hits	3	3

postsummary2 32 ×

Output

Challenge 2 — Trigger

Trigger on likes table:

When a user likes a post, insert a row into a new table:

notifications (userid, message, createdat)

```
629
630 •  select * from notifications;
631
632
633
```

Result Grid | Filter Rows: | Edit: | Export/Imp

NotificationID	UserID	Message	CreatedAt
1	101	You liked Post ID 2	2026-10-30 19:55:26
2	102	You liked Post ID 3	2026-10-30 19:55:26
3	105	You liked Post ID 4	2026-10-30 19:55:26
NULL	NULL	NULL	NULL

notifications 35 ×

Output